

KARTIKEY GUPTA

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PROFESSIONAL EXPERIENCE

AI & Machine Learning Intern | AMITKK Digital Solutions

June 2026 – July 2026

- Built a **Python lead-generation and outreach platform** (PyQt5 desktop client + **FastAPI** backend) that collects business and professional data using **DuckDuckGo search and Selenium**, scraping **~80 LinkedIn posts and ~100 websites per day**, with MongoDB for project-based storage.
- Integrated the **Gemini API** to generate personalised outreach **emails and social comments** for each lead, **triggered automatically the moment new data lands in the database**, with a **human-in-the-loop approval step** before anything is sent.
- Implemented a **scheduler-triggered** scraping workflow and **automated email dispatch to all scraped leads every 30 minutes**.
- Designed the **database architecture** (project-level data association) and added **deduplication, input validation, and website/domain filtering** to improve data quality.
- Debugged **duplicate records, scheduler execution, and API/database synchronisation** issues across the scraping and email pipeline.

EDUCATION

Bharati Vidyapeeth's College of Engineering, Delhi

2023 – 2027

Bachelor of Technology in Computer Science | CGPA: 8.74/10

TECHNICAL SKILLS

Programming Languages: Python, JavaScript

Backend & Agentic AI: FastAPI, FastMCP, Gemini API, Tool-calling, RAG, Structured Output, Pydantic, Selenium, PyQt5

Frameworks & Libraries: Scikit-learn, TensorFlow, Pandas, Matplotlib, Node.js, React.js

Databases & Tools: MongoDB, MySQL, ChromaDB, Git/GitHub, JWT, Docker, pytest

PROJECTS

CareLink – Hospital Discharge & Follow-Up Assistant | Python, FastAPI, FastMCP, Gemini API, RAG, ChromaDB

- Built a **multi-turn conversational agent** (custom tool-calling loop) behind a **FastAPI** webhook that answers patient questions on discharge instructions and books follow-ups, escalating to staff when unsure.
- Exposed **5 tools via a FastMCP server** (patient record lookup, medication schedule, appointment booking, reminder scheduling, staff escalation) with **Pydantic-validated structured output**.
- Implemented RAG over discharge summaries and care guidelines using ChromaDB embeddings.
- Wrote a safety test harness covering medical-advice requests, missed medication, emergency symptoms, and invalid input, with pass/fail checks.

TaskChain – Role-Based Workflow API | Python, FastAPI, MongoDB, JWT, Pydantic, Docker

- Built a **scalable FastAPI backend** supporting multi-level organisational task workflows with approval chains.
- Implemented **JWT authentication and role-based access control** for Admin, Head, and Member levels.
- Designed **10+ async REST APIs** with Pydantic request validation, pagination, and auto-generated OpenAPI docs.
- Added indexed MongoDB queries and a pytest suite covering the API endpoints; containerised with Docker.

Kitchen Preparation Time Prediction System | Python, Scikit-learn, FastAPI, React.js

- Built an ML pipeline to predict kitchen preparation time from operational restaurant data using a **two-stage Random Forest** architecture.
- Engineered behavioural and congestion-related restaurant features; **reduced MAE from 3.50 to 0.31 minutes and P90 error from 5.63 to 0.84 minutes**.
- Served the model through a **FastAPI prediction endpoint** and built a **React.js dashboard** giving delivery platforms live prep-time estimates per order.

ACHIEVEMENTS

- Co-authored and presented a research paper, “**A Systematisation of Adversarial Multi-Agent Learning in Cybersecurity for Web Security**”, at AKGEC 2026 (3rd International Conference on Green Energy and Computing). [Paper](#)
- Organised **BrainBot Ideathon** (CodeChef, BVCOE) and **CodeJam Talks** (ACM-BVP).